

Fig. 3: Actual-size PCB layout of water refiller for air-coolers

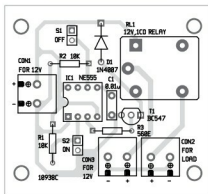


Fig. 4: Components layout for the PCB

using a small speaker magnet and a hollow float ball. Diameter of the magnetic float should be slightly less than the PVC pipe's so that it can float freely from bottom to top, or vice versa, based on the water level. If possible, fit the pipe in the tank's inner wall with reed switches affixed externally.

Note. If the load is a 12V solenoid valve as shown in Fig. 5, use 12V DC supply to power the circuit and the load. Alternatively, use 230V AC



Fig. 5: 12V solenoid valve

PARTS LIST

Semiconductors:

- IC1 - NE555 timer
- T1 - BC547 npn transistor
- D1 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R2 - 10-kilo-ohm
- R3 - 560-ohm

Capacitors:

- C1 - 0.01 μ F ceramic disk

Miscellaneous:

- CON1-CON3 - 2-pin terminal connector
- S1, S2 - Reed switch normally open type
- RL1 - 12V, single-changeover relay
- Small float magnet or small speaker magnet and a hollow float ball
- 12V adaptor or 12V battery
- 12V solenoid valve or 230V operated centrifugal pump

supply at CON3 and external 12V adaptor at CON1, if the load is a 230V centrifugal pump. **EFY**



A. Samiuddin is B.Tech in electrical and electronics engineering. His interests include LED lighting, power electronics, microcontrollers and Arduino programming



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